A nilpotent Freiman dimension lemma

Terence Tao · 21 December 2011

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Emmanuel Breuillard, Ben Green and I have just uploaded to the arXiv the short paper "A nilpotent Freiman dimension lemma", submitted to the special volume of the European Journal of Combinatorics in honour of Yahya Ould Hamidoune. This paper is a nonabelian (or more precisely, nilpotent) variant of the following additive combinatorics lemma of Freiman:

> **Freiman's lemma.** Let A be a finite subset of a Euclidean space with $|A+A| \leq K|A|$. Then A is contained in an affine subspace of dimension at most $\lfloor K-1 \rfloor$.

This can be viewed as a "cheap" version of the more well known theorem of Freiman that places sets of small doubling in a torsion-free abelian group inside a generalised arithmetic progression. The advantage here is that the bound on the dimension is extremely explicit.

Our main result is

> **Theorem.** Let A be a finite subset of a simply-connected nilpotent Lie group G which is a K-approximate group (i.e. A is symmetric, contains the identity, and $A \cdot A$ can be covered by up to K left translates of A. Then A can be covered by at most K^{2+29K^9} left-translates of a closed connected Lie subgroup of dimension at most K^9.

We remark that our previous paper established a similar result, in which the dimension bound was improved to $6 \log_2 K$, but at the cost of worsening the covering number to $O_K(1)$, and with a much more complicated proof (91 pages instead of 8). Furthermore, the bound on $O_K(1)$ is ineffective, due to the use of ultraproducts in the argument (though it is likely that some extremely lousy explicit bound could eventually be squeezed out of the argument by finitising everything). Note that the step of the ambient nilpotent group G does not influence the final bounds in the theorem, although we do of course need this step to be finite. A simple quotienting argument allows one to deduce a corollary of the above theorem in which the ambient group is assumed to be residually torsion-free nilpotent instead of being a simply connected nilpotent Lie group, but we omit the statement of this corollary here.

To motivate the proof of this theorem, let us first show a simple case of an argument of Gleason, which is very much in the spirit of Freiman's lemma:

> **Gleason Lemma (special case).** Let A be a finite symmetric subset of a Euclidean space, and let $0 = H_0 \subset H_1 \subset \ldots \subset H_k$ be a sequence of subspaces in this space, such that the sets $2A \cap H_i$ are strictly increasing in i for $i = 0, \ldots, k$. Then $|5A| \geq k|A|$, where $5A = A+A+A+A+A$.

Proof. By hypothesis, for each $i = 1, \ldots, k$, the projection B_i of $2A \cap H_i$ to H_i/H_{i-1} is non-trivial, finite, and symmetric. In particular, since the vector space H_i/H_{i-1} is torsion-free, $B_i + B_i$ is strictly larger than B_i. Equivalently, one can find a_i in $(2A \cap H_i) + (2A \cap H_i)$ that does not lie in $(2A \cap H_i) + H_{i-1}$; in particular, $a_i \in 4A \cap H_i$ and $a_i + A$ is disjoint from $H_{i-1} + A$. As a consequence, the $a_i + A$ are disjoint and lie in 5A, whence the claim. □

Note that by combining the contrapositive of this lemma with a greedy algorithm, one can show that any K-approximate group in a Euclidean space is contained in a subspace of dimension at most K^4, which is a weak version of Freiman's lemma.

To extend the argument to the nilpotent setting we use the following idea. Observe that any non-trivial genuine subgroup H of a nilpotent group G will contain at least one non-trivial central element; indeed, by intersecting H with the lower central series $G = G_1 \geq G_2 \geq G_3 \geq \ldots$ of G, and considering

the last intersection $H \cap G_k$ which is non-trivial, one obtains the claim. It turns out that one can adapt this argument to approximate groups, so that any sufficiently large K-approximate subgroup A of G will contain a non-trivial element that centralises a large fraction of A. Passing to this large fraction and quotienting out the central element, we obtain a new approximate group. If, after a bounded number of steps, this procedure gives an approximate group of bounded size, we are basically done. If, however, the process continues, then by using some Lie group theory, one can find a long sequence $H_0 \leq H_1 \leq \ldots \leq H_k$ of connected Lie subgroups of G, such that the sets $A^2 \cap H_i$ are strictly increasing in i. Using some Lie group theory and the hypotheses on G, one can deduce that the group $\langle A^2 \cap H_{i+1} \rangle$ generated by $A^2 \cap H_{i+1}$ is much larger than $\langle A^2 \cap H_i \rangle$, in the sense that the latter group has infinite index in the former. It then turns out that the Gleason argument mentioned above can be adapted to this setting.

Comments (1)



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[Added, thanks – T.]